

AMEYA PRATAP SINGH

B.E. Computer Science Engineering · BMSCE Bengaluru

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EDUCATION

BMS College of Engineering, Bengaluru

B.E. – Computer Science & Engineering

2024 – Present

CGPA: 9.57 / 10 (3 Semesters)

New Era High School

Class XII – CBSE

2021 – 2022

Score: 95.6%

New Era High School

Class X – CBSE

2019 – 2020

Score: 95.8%

TECHNICAL SKILLS & COURSEWORK

Languages: C, Java, Python, JavaScript, TypeScript

Web & Frameworks: HTML, CSS, React, Next.js, Node.js, Express.js, Tailwind CSS

Databases & ORMs: MySQL, MongoDB, PostgreSQL, Prisma

Tools & Platforms: Git & GitHub, VS Code, Vercel, Postman

Core Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Organisation & Architecture, Operating Systems, Probability & Statistics, Linear Algebra, Calculus

PROJECTS

SubTrack — Subscription Management Platform [GitHub ↗](#) [Live ↗](#)

Next.js · TypeScript · PostgreSQL ·
Prisma · Auth.js

- Built a full-stack subscription management platform that tracks recurring billing, sends automated email reminders, and visualises spending trends via a Recharts analytics dashboard.
- Implemented Google OAuth via Auth.js v5, server-side data fetching with React Server Components, and mutation handling through Next.js Server Actions — eliminating the need for client-side API routes.
- Designed a cron-driven email system (Nodemailer/Gmail SMTP, Vercel Cron Jobs) with HMAC-derived unsubscribe tokens and duplicate-send prevention using a lastNotifiedAt deduplication field.
- Extended V2 with service logo auto-detection (Clearbit → favicon fallback), CSV export, budget alert banners, and shared subscription splitting with per-person cost propagation.

Vendora — Full-Stack E-Commerce Platform [GitHub ↗](#) [Live ↗](#)

React · Node.js · Express · MongoDB
· Razorpay

- Developed a premium MERN-stack boutique e-commerce application with JWT authentication, product search & category filtering, a cart/wishlist system, and Razorpay payment integration.
- Built a full admin dashboard with CRUD product management, Cloudinary image hosting with automatic orphaned-image cleanup, and paginated order management.
- Implemented secure bcrypt password hashing, HMAC-SHA256 Razorpay signature verification, and server-side pagination for performant large-catalog browsing.

- Built a full-stack developer tool that clones any public GitHub/GitLab/Bitbucket repository, runs AST extraction via graphify to build a cross-file dependency graph, then feeds 1-hop call/import context alongside each file into an LLM to generate structural, flow-aware code explanations anchored inline to the source.
- Implemented a dual-provider LLM backend with an automatic model waterfall — NVIDIA NIM (llama-3.1-70b-instruct) as primary and Google Gemini (gemini-2.5-flash / gemini-2.5-pro) as fallback — across both /explain and /chat endpoints, with a fast/smart tier selector on the frontend mapping to different models per provider.
- Designed a 4-panel resizable workspace (Monaco editor, AST-driven explanation panel, file tree, Deep Dive chat tray) where users can highlight any passage to trigger a context-scoped conversational chat — with full conversation history, session-cached explanations, and the complete repo file tree injected into the system prompt for structure-aware answers.